Fibonacci-Type Sequences

AQA · KS3 Year 7–8

Answer each question in your workbook. Show your working.

Method

Each term = sum of the two before it

Continue by adding the last two

Missing term: use term = prev + prev-prev

Work backwards by subtracting

Check the rule everywhere

Common mistakes

- Multiplying instead of adding.
- Adding the wrong pair of terms.
- Missing term: term = (term before) + (two before).
- Arithmetic slips with larger numbers.

Worked example

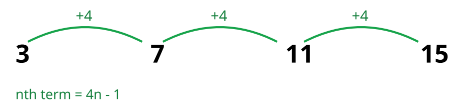
Continue 1, 1, 2, 3, 5, 8, ...

Next = 5 + 8 = 13, then 8 + 13 = 21.

Backwards check: 13 - 8 = 5

Visual aid

Visual Example



Each term is built from the two before it: 1, 1, 2, 3, 5, 8, 13, ... This is the famous Fibonacci pattern, and the ratio of neighbouring terms approaches the golden ratio (1.618).

Questions

Answer each question in your workbook. Show your working.

Q1 of 8 **[2]** ●○○○○

Write the next two terms: a) 1, 1, 2, 3, 5, 8, … b) 2, 2, 4, 6, 10, …

Q2 of 8 **[2]** ●○○○○

Each term = sum of the two before. Write the next term: a) 3, 5, 8, … b) 1, 4, 5, 9, …

Q3 of 8 **[3]** ●●○○○

Find the missing term: a) 1, __, 3, 5, … b) 4, __, 11, 18, … c) __, 5, 8, 13, …

Q4 of 8 **[3]** ●●○○○

Write the first five terms: a) start 1, 3 b) start 2, 5 c) start 4, 4

Q5 of 8 **[3]** ●●●○○

a) First two terms 1 and 2 — write the first six terms. b) 3rd term 10, 4th term 16 — find the 1st and 2nd terms.

Q6 of 8 **[3]** ●●●○○

Rabbit pairs: months 1 and 2 each have 1 pair; each month = sum of previous two. a) Pairs in months 3–6. b) First month with more than 10 pairs?

Q7 of 8 **[4]** ●●●●○

3rd term 7, 4th term 11. a) Find 1st and 2nd terms. b) First six terms. c) 7th term.

Q8 of 8 **[5]** ●●●●●

5th term 22, 6th term 35. a) Work backwards to first four terms. b) Whole sequence (1–6). c) Why does it grow more like a geometric sequence?